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Volume 05 — Diffusion (L44–L53)

Generative AI & LLMs · NPTEL · Generative AI / LLMs **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

L44: Introduction to diffusion models

Learning focus

- Place diffusion after autoencoders, VAEs, and GANs on the course timeline, including the **stability vs. quality** gap it was meant to close.
- Read a **probability landscape**: high-p peaks = realistic images, low-p valleys = noise, and why a model needs a **compass** (local next-step choices).
- Contrast a **fixed forward** noising chain with a **learned reverse** denoising chain, and sketch training vs. inference.
- State the **Markov** property used for both chains (snakes-and-ladders example).

Exam lines

- Diffusion = many easy Markov steps instead of one intractable map from noise to data.
- Forward is **fixed** Gaussian corruption; reverse is the **learned** denoiser.
- High probability ↔ structure; low probability ↔ noise; generation is climbing with a local rule $p_t(x)$.
- The net is trained to predict **noise**, then used at inference to denoise $x_T \sim \mathcal{N}(0, I)$ down to an image.

Concept tables

Year (as taught)	Model	What it did	Limitation named here
1987	Autoencoder	Compress to a latent, **reconstruct** the input (zip / unzip a folder)	**Cannot** emit realistic **new** images; used for denoising, anomaly spotting, feature extraction, scratch restoration
2013	VAE	Latent is a **Gaussian** (μ, σ), not a single point	**Can** make variants (ship with different sky / water / hull color) but images are often **blurry**, not high-res
2015	GAN	High-quality fakes from **noise**; FaceApp (2017) and This Person Does Not Exist as GAN apps	**Mode collapse** (train on dogs/cats/horses/birds, emit only dogs) and **unstable** training
2015	Diffusion	Physics-style **diffusion** of probability	Designed to be **as stable as VAEs** and **as sharp as GANs**

Idea	Why it fails
——	———— —-
Map **every image in the world** as a peak	Impossible
Memorize **known** rooms of “your house”	No help in an **un-known house** (new images)

L45: Mathematics of diffusion models

Learning focus

- Write the **forward** kernel $q(x_t \mid x_{t-1})$ and the **reverse** kernel $p_\theta(x_{t-1} \mid x_t)$ as the lecture set them up.
- Explain the **noise variance schedule** β_t and what goes wrong if destruction is too fast or too slow.
- Recall the **Gaussian** PDF, the **standard** Gaussian $\mathcal{N}(0, 1)$, and why images need a **multivariate** Gaussian (covariance, not a scalar σ^2).
- Compute a 2×2 **covariance matrix** on the height–weight table (population n, not $n - 1$).

Exam lines

- q noising is predefined Gaussian; p_θ denoising is the trainable Gaussian.
- β_t is a **schedule**, not a constant, so x_T is fully noise without making reverse learning impossible.
- Standard Gaussian = $\mathcal{N}(0, 1)$; pixels need $\mathcal{N}(\mu, \Sigma)$.
- Σ's diagonal is variance, off-diagonal is relationship; the $n = 4$ height–weight example is the calculation you are expected to reproduce.

Concept tables

Symbol	Meaning
x_0	original data (clean image / signal)
x_t	noisy version at step t
T	total diffusion steps (can be **up to ~1000**, application-dependent)
$t \in \{1, \dots, T\}$	current step
–	Forward ———
Direction	clean → noise
Learning	none
Typical letter	q

L46: DDPM — forward process

Learning focus

- Identify **DDPM** as the 2020 Ho et al. (UC Berkeley) formulation of diffusion.
- Write the Markov forward kernel $q(x_t \mid x_{t-1})$ with mean $\sqrt{1-\beta_t}\, x_{t-1}$ and covariance $\beta_t I$.
- Show why the previous image must be **scaled** (unscaled variance **blows up**).
- Expand the recursion to the **closed-form** $q(x_t \mid x_0)$ and the **reparameterization** that samples any t in one shot.

Exam lines

- DDPM forward is a **fixed** Markov Gaussian: $q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I)$.
- $\alpha_t = 1 - \beta_t$; samples $x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \varepsilon_t$.
- Skipping the $\sqrt{\alpha_t}$ scale makes variance **accumulate** until the law blows up.
- $\bar{\alpha}_t = \prod_s \alpha_s$ yields $q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t) I)$ and the one-shot $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$.

Concept tables

Forward	Reverse
Clean image → almost pure noise, step by step	Almost pure noise → clean image
No learning; predefined Gaussian	Learning lives here (next lecture)

Symbol	Meaning
x_t	image at step t
x_{t-1}	image at step $t-1$
β_t	how much noise is added at t (magnitude)
I	identity matrix

L47: DDPM — reverse process

Learning focus

- Write the reverse **Markov** joint $p_\theta(x_{0:T})$ from a Gaussian **prior** x_T.
- State the **forward posterior** $q(x_{t-1} \mid x_t, x_0)$ (mean $\tilde{\mu}_t$ depends on x_0 — unusable at generation).
- Eliminate x_0 using the forward reparameterization, replace ε by ε_θ, and sample x_{t-1}.
- Quote the **MSE noise-prediction** loss used to train DDPM.

Exam lines

- Reverse joint = prior $x_T \sim \mathcal{N}(0, I)$ times Markov conditionals $p_\theta(x_{t-1} \mid x_t)$.
- $q(x_{t-1} \mid x_t, x_0)$ is Gaussian, but $\tilde{\mu}_t$ needs the **unknown** x_0 at sampling time.
- $\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t))$.
- $x_{t-1} = \mu_\theta + \sigma_t z$; train ε_θ with MSE vs true ε.

Concept tables

Factor	Meaning
$p_\theta(x_{0:T})$	joint of the **entire reverse trajectory**
$p(x_T)$	**prior** at the start of generation
$\prod p_\theta(x_{t-1} \mid x_t)$	all reverse conditionals
θ	trainable weights

Term	Lecture meaning
Prior	what you assume **before** extra evidence
Posterior	the law **after** observing more information

L48: Hands-on — forward diffusion process

Learning focus

- Implement the DDPM **forward** kernel, α_t, $\bar{\alpha}_t$, and **closed-form** x_t from x_0.
- Compare **linear** vs **cosine** β-schedules ($T = 1000$) on remaining signal and **SNR**.
- Show that **iterative Markov** noising and **one-shot** $q(x_t \mid x_0)$ match in **mean and std** (Monte Carlo).
- Train a small **time-conditioned CNN** on MNIST to predict ε, then reconstruct by **subtracting** predicted noise — and list what a full DDPM still needs.

Exam lines

- Code $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$ and the Markov one-step kernel; they match in Monte Carlo.
- $T = 1000$; cosine β beats linear on uniform signal decay and final SNR.
- Astronaut viz: linear is "done" by ~500; cosine fades evenly.
- MNIST CNN (~124k params, 12k/2k split) can drop L_{simple} and roughly undo noise in one subtract — that is **not** yet a DDPM sampler.

Concept tables

Process	Path	Who learns
Forward	$x_0 \to x_1 \to \cdots \to x_T$	nobody — add small Gaussians until noise
Reverse	$x_T \sim \mathcal{N}(0, I)$ toward x_0	**neural net**

Schedule	Shape on the plot	Remaining signal $\bar{\alpha}_t$
Linear	essentially **flat** $\beta_t \approx 1/1000$	signal decays **slowly at first**, then **drops**
Cosine	β_t **small at the start**, then **rises**	**more uniform** decline of original signal

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L49: U-Net for denoising

Learning focus

- Place **U-Net** in DDPM: input (x_t, t) → predicted noise $\varepsilon_\theta(x_t, t)$.
- Contrast **Ronneberger** segmentation U-Net with the DDPM stack: **time embeddings**, **residual blocks**, **self-attention**.
- Compute a **sinusoidal** time embedding (the $T = 100$, $d = 8$ numerical example).
- Walk **encoder / bottleneck / decoder**, the residual-block algebra (GroupNorm, **SiLU**, time projection), and **QKV** attention.

Exam lines

- U-Net is the DDPM denoiser: global structure + skips for fine detail.
- TE(t) uses sin / cos of $t/10000^{2i/d}$; the $t = 100$, $d = 8$ vector is the exam-style drill.
- Residual path: GroupNorm → SiLU → conv → **add** W_tTE(t) → GroupNorm → SiLU → conv → $H + F(H)$.
- Attention: Q, K, V projections, softmax$(QK^\top/\sqrt{d})$, then AV, so far-apart structure (two eyes) can interact.

Concept tables

Part	Job
Encoder (contracting)	downsample; spatial size shrinks; keep important information
Bottleneck	most **compressed** summary
Decoder (expanding)	upsample / reconstruct
Skips	**crop and concatenate** encoder features onto the matching decoder level so fine detail survives

Example t	Typical look (as taught)
50	slightly noisy
500	moderately noisy
900	heavily noised

L50: Classifier-guided diffusion model

Learning focus

- Explain why an unconditional DDPM cannot choose a class (dog vs cat vs flower).
- Describe the **artist + critic** pair: a denoising U-Net plus a **noisy-image classifier**.
- Reconstruct the Bayes / score derivation that turns a classifier gradient into a **guided noise** $\hat{\varepsilon}$.
- State the **guidance scale** s trade-off (class precision vs diversity) and the sampling replacement rule.
- List the pipeline limitations that motivate classifier-free guidance in the next lecture.

Exam lines

- Unconditional DDPM generates realistic images with **no class knob**.
- Classifier guidance adds a noisy-image critic; $\nabla_{x_t} \log p_\phi(y \mid x_t)$ steers denoising toward y.
- Bayes splits the conditional score into a **realism** term (U-Net ε_θ) and a **class** term (classifier).
- Guided noise $\hat{\varepsilon}$ replaces ε_θ in the reverse update; s trades **precision vs diversity**.

- Cost: a second network, noisy-label training, per-step gradients, and adversarial artifacts. Next lecture: **classifier-free** guidance.

Concept tables

Role	What it is	What it does
Artist	Ordinary **unconditional** DDPM / U-Net	Denoises a noisy image toward something realistic. It does **not** know which class to emit.
Critic	A **classifier**, itself a pretrained network	Looks at the noisy image x_t and predicts $p_\phi(y \mid x_t)$ over classes (cats, dogs, trees, houses, ...).
s	Class recognizability	Sample diversity
—–	————————	———————
Small	Weak conditioning; images may not look like the target class	Higher diversity
Large	Strong, precise class (very "dog-like")	Diversity **drops**

L51: Classifier-free diffusion models

Learning focus

- Restate why classifier-guided diffusion is heavy (extra noisy classifier, no off-the-shelf ResNet).
- Explain **classifier-free guidance (CFG)**: one U-Net trained for **both** conditional and unconditional denoising.
- Write the CFG noise combination with guidance scale s.
- Contrast sampling cost: U-Net **+ classifier** vs **two U-Net passes**.

Exam lines

- CFG exists to **delete the noisy classifier** while keeping class control.
- One U-Net, **joint** training: keep y or replace it by null ϕ with fixed dropout probability.
- The old classifier gradient is rewritten as **conditional minus unconditional** diffusion scores.
- $\hat{\varepsilon} = (1+s)\varepsilon_\theta(x_t, t, y) - s\varepsilon_\theta(x_t, t, \phi)$; larger s (paper demo $s = 3$) sharpens class at some cost (saturation / diversity).
- Sampling trades classifier cost for **two U-Net evaluations**. Next lecture: **Stable Diffusion**.

Concept tables

Problem	Why it hurts
Extra classifier	Must be trained **on noisy images** x_t to supply $\nabla_{x_t} \log p(y \mid x_t)$
No ImageNet ResNet	Off-the-shelf nets saw **clean** pixels, not x_t
Two trainings	Unconditional DDPM **and** the classifier → slower, more complex pipeline
	Classifier-guided (Lec 50)
–	——————————-
Separate classifier	Yes
Noisy-classifier training	Yes
Where the guidance signal comes from	Classifier gradient $\nabla_{x_t} \log p_\phi(y \mid x_t)$
Sampling-time cost	U-Net **plus** classifier, both run while generating

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L52: Stable Diffusion model

Learning focus

- Motivate **latent** diffusion: why pixel-space DDPM on $512 \times 512 \times 3$ is too expensive.
- Separate **latent diffusion (LDM)** (the algorithm) from **Stable Diffusion** (text-to-image implementation).
- Trace the frozen autoencoder → noisy latent → U-Net → decoder pipeline and the latent-space MSE objective.
- Write **cross-attention** with Q from U-Net maps and K, V from a **CLIP** text encoder.
- Choose **concatenation vs cross-attention** from whether the condition has spatial structure.

Exam lines

- LDM moves DDPM from pixels into a **frozen autoencoder's latent** ($64 \times 64 \times 4$ vs $512 \times 512 \times 3$).
- Stable Diffusion **is** that algorithm plus **CLIP text embeddings** and **cross-attention** into the U-Net.
- Loss is DDPM's noise MSE on z_t, not x_t.
- Q from U-Net maps, K and V from $\tau_\theta(y)$; concat is for **spatial** conditions.
- Next in the playlist is the **reverse-diffusion lab**; the instructor also flags the coming shift to **NLP / transformers / LLMs**.

Concept tables

Name	What it is
Latent diffusion model (LDM)	The **algorithm** in that paper

Name	What it is
Stable Diffusion	A **practical** LDM implementation for **text-to-image**, with extra **conditioning**
Symbol	Meaning
——–	———
ε	Gaussian noise added in the **forward** process
z_t	**Noisy latent** at time t
$\varepsilon_\theta(z_t, t)$	U-Net prediction — depends on z_t **and** t, **not** on pixel x_t

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L53: Hands-on reverse diffusion process

Learning focus

- Implement reverse DDPM sampling from $\mathcal{N}(0, I)$ with a **small class-conditional U-Net** on **MNIST**.
- Train with L_{simple} (MSE on Gaussian noise) and **null-class dropout** for classifier-free guidance.
- Compare **guidance scales** and a tiny **quality classifier** on generated digits.
- Contrast diffusion vs **GAN** vs **VAE** using the lecture's table (tendencies, not universal laws).

Exam lines

- Reverse diffusion repeatedly maps noise toward data; a U-Net mixes global structure with skip-connection detail.
- Train by predicting known Gaussian noise with **MSE**; condition on t (noise level) and optionally y.
- Null-class dropout + $\varepsilon_{\text{uncond}} + s(\varepsilon_{\text{cond}} - \varepsilon_{\text{uncond}})$ is **classifier-free** guidance. Moderate s helps class adherence; excessive s hurts diversity / adds artifacts.
- This classroom model **demonstrates the mechanism**; scale data, T, channels, and epochs to get convincing digits. That closes the diffusion hands-on pair (forward + reverse).

Concept tables

Setting	Value in the notebook
Time steps T	**100** (`fast_mode=True`; set `False` to increase T)
β schedule	**Linear**, β from **0** to **0.02**
α_t	$1 - \beta_t$
Also computed	$\bar{\alpha}_t$, posterior variance
Split	Size
——-	——
Train	**12,000**
Validation	**2,000**